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1. Homologous Series: Definition & Characteristics

- Homologous series = same general formula aur same functional group wale compounds; successive members mein sirf carbon chain length alag hoti hai. [Q532, Q533, Q535, Q537, Q601]
- Har successive member mein CH_2 (methylene) group ka difference hota hai; molecular mass difference = 14 u ($\text{C}=12 + \text{H}_2=2$). [Q532, Q605]
- Homologous series mein Hydrogen atom ko hi different functional group se replace kiya jaata hai. [Q575]
- Same homologous series ke members ki chemical properties nearly identical hoti hain; physical properties gradually change — molecular mass badhne ke saath boiling point bhi badhta hai. [Q532, Q569]
- Examples: Alkanes (CH_4 , C_2H_6 , C_3H_8 ...), Alkenes (C_2H_4 , C_3H_6 , C_4H_8 ...), Alcohols (CH_3OH , $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_3\text{H}_7\text{OH}$...). [Q533, Q535]
- CH_4 , C_2H_6 , C_3H_6 same homologous series mein nahi hain — C_3H_6 ek alkene hai, alkane nahi. [Q542]
- C_4H_8 (alkene) baaki alkanes ke saath same homologous series mein nahi aata. [Q555]
- $\text{C}_4\text{H}_{10}\text{OH}$ ek valid alcohol formula nahi hai — isliye alcohols ki homologous series mein nahi aata. [Q567]
- Alkanes mein sabse zyada melting point wala member options mein C_4H_{10} (Butane) hai. [Q551]
- Homologous series mein sabse kam boiling point wala member generally sabse chhota carbon chain wala hota hai — jaise CH_4O (Methanol). [Q562]
- Boiling point increasing order example: $\text{CH}_4 < \text{CHCl}_3 < \text{CH}_3\text{CH}_2\text{OH} < \text{CH}_3\text{COOH}$. [Q564]

2. Hydrocarbons: Alkanes, Alkenes, Alkynes & Cycloalkanes

- Alkanes (Saturated): C-C single bonds; general formula C_nH_{2n+2} ; examples: Methane CH_4 , Ethane C_2H_6 , Propane C_3H_8 , Butane C_4H_{10} , Pentane C_5H_{12} . [Q534, Q536, Q538]
- Ethane mein har carbon atom 4 other atoms se bonded hota hai (3H + 1C, ya symmetric structure mein 4 atoms total bonds); Ethane mein sirf ek hi C-C single bond hota hai. [Q619, Q577]
- Alkenes (Unsaturated): C=C double bond; general formula C_nH_{2n} ; examples: Ethene C_2H_4 , Propene C_3H_6 (alkene series ka second member), Butene C_4H_8 . [Q536, Q538, Q571]
- Propene ek double bond wala compound hai. [Q628]
- Alkynes (Unsaturated): $C\equiv C$ triple bond; general formula C_nH_{2n-2} ; examples: Ethyne C_2H_2 , Propyne C_3H_4 , Butyne C_4H_6 . [Q538, Q540]
- Bond length order (increasing): Ethyne (triple bond) < Ethene (double bond) < Ethane (single bond) — jitne zyada bonds utni chhoti bond length. [Q609]
- Cycloalkanes: Ring structure; saturated; formula C_nH_{2n} ; example: Cyclohexane C_6H_{12} (6 C-C bonds + 12 C-H bonds = 18 total covalent bonds); C_6H_{12} formula cyclic hydrocarbons se related hai. [Q449, Q540, Q557]
- Cyclopentane mein total 15 single bonds hote hain (5 C-C ring bonds + 10 C-H bonds). [Q546]
- First members: Alkane = Methane (CH_4); Alkene = Ethene (C_2H_4); Alkyne = Ethyne (C_2H_2). [Q534, Q536, Q538]
- Benzene (C_6H_6) mein 3 double bonds hote hain (resonance structure). [Q573, Q614]

3. Functional Groups & IUPAC Nomenclature

- Functional group = atom/group jo molecule ki chemical properties decide karta hai. [Q541, Q543, Q545]
- Common functional groups: $-OH$ (Alcohol, suffix -ol, e.g. C_2H_5OH Ethanol); $-CHO$ (Aldehyde, suffix -al, e.g. $HCHO$ Methanal — aldehyde group ka symbol hai); $>C=O$ (Ketone, suffix -one, e.g. CH_3COCH_3 Propanone; R-CO-R ketone represent karta hai); $-COOH$ (Carboxylic acid, suffix -oic acid, e.g. CH_3COOH Ethanoic acid); $-COOR$ (Ester); $-X$ (Halide, prefix chloro-, e.g. CH_3Cl); $-NH_2$ (Amine, suffix -amine); $-O-$ (Ether, e.g. CH_3OCH_3 Dimethyl ether). [Q541, Q543, Q545, Q547, Q589, Q548, Q623]
- Carbon count prefixes: meth-(1), eth-(2), prop-(3), but-(4, jaise 4-carbon hydrocarbon ka prefix "But" hota hai), pent-(5), hex-(6), hept-(7), oct-(8), non-(9), dec-(10). [Q549, Q603]
- Common names: Ethanol = ethyl alcohol; Ethanoic acid = acetic acid; Propanone = acetone. [Q547, Q549]
- Structure-name pairs: C_6H_6 = Benzene; $H_2C=CH_2$ = Ethylene/Ethene; CH_3-CHO = Acetaldehyde; CH_3COCH_3 = Acetone; CH_3CH_3 = Ethane. [Q544]

4. Isomerism in Organic Compounds

- Isomers = same molecular formula, different structural formula/arrangement — inhe hydrocarbons ke context mein bhi "isomers" hi kehte hain. [Q550, Q552, Q554, Q633]
- Chain isomerism: different carbon skeleton — n-Butane vs Isobutane. [Q552]
- Position isomerism: same functional group, different position — Propan-1-ol vs Propan-2-ol. [Q554]
- Functional group isomerism: different functional groups — Ethanol (C_2H_5OH) vs Dimethyl ether (CH_3OCH_3), both C_2H_6O . [Q554]
- Butane (C_4H_{10}) ke 2 isomers; Pentane (C_5H_{12}) ke 3 isomers. [Q552, Q554]

5. Combustion & Flame Characteristics

- All hydrocarbons air/oxygen mein burn karke $CO_2 + H_2O + \text{Heat}$ dete hain. [Q556, Q558, Q560]

- Saturated hydrocarbons (Alkanes) = clean blue flame; sufficient oxygen mein complete combustion: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$. [Q556, Q558]
- Unsaturated hydrocarbons (Alkenes/Alkynes) = sooty yellow flame; kam hydrogen content ki wajah se incomplete combustion hoti hai. [Q558, Q560]
- Alkynes sabse zyada sooty flame dete hain kyunki inka carbon:hydrogen ratio sabse zyada hota hai; Benzene bhi sooty flame deta hai. [Q560]

6. Addition Reactions of Unsaturated Compounds

- Addition reaction = unsaturated compound mein atoms/groups add hote hain double/triple bond break hokar; sirf alkenes aur alkynes mein hoti hai. [Q561, Q563, Q565]
- Hydrogenation: H_2 add karke saturated compound banana; vegetable oil $\rightarrow$ Vanaspati ghee, Nickel (Ni) catalyst use hota hai. $\text{C}_2\text{H}_4 + \text{H}_2 \rightarrow \text{C}_2\text{H}_6$. [Q561, Q563, Q612, Q639]
- Halogenation: Cl_2 ya Br_2 add karna; bromine water decolourise — unsaturation ka test. $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2$. [Q563, Q565]
- Hydration: H_2O add karke alcohol banana. $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightarrow \text{C}_2\text{H}_5\text{OH}$ (conc. H_2SO_4 catalyst, 300°C , 60 atm). [Q565]
- Addition reaction alkanes mein nahi hoti. [Q561]

7. Substitution Reactions of Saturated Compounds

- Substitution reaction = saturated compound mein ek atom/group replace hota hai dusre atom/group se; alkanes mein hoti hai. [Q566, Q568, Q570]
- Chlorination of methane (UV light): $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (Methyl chloride). $\text{CH}_3\text{Cl} + \text{Cl}_2 \rightarrow \text{CH}_2\text{Cl}_2 + \text{HCl}$ (Methylene chloride). $\text{CH}_2\text{Cl}_2 + \text{Cl}_2 \rightarrow \text{CHCl}_3 + \text{HCl}$ (Chloroform). $\text{CHCl}_3 + \text{Cl}_2 \rightarrow \text{CCl}_4 + \text{HCl}$ (Carbon tetrachloride). [Q566, Q568]
- UV light/heat ki zaroorat hoti hai substitution ke liye. [Q568]

8. Alcohols: Classification, Properties & Oxidation

- Primary alcohol (1°): $-\text{OH}$ terminal carbon par; $-\text{CH}_2\text{OH}$; example: Ethanol $\text{CH}_3\text{CH}_2\text{OH}$. [Q570, Q572]
- Secondary alcohol (2°): $-\text{OH}$ middle carbon par; $>\text{CHOH}$; example: Propan-2-ol $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$. [Q572]
- Tertiary alcohol (3°): $-\text{OH}$ carbon par jo 3 other carbons se jura ho; example: 2-Methylpropan-2-ol. [Q572]
- Alcohols ka oxidation: Primary alcohol $\rightarrow$ Aldehyde $\rightarrow$ Carboxylic acid; ethanol $\text{K}_2\text{Cr}_2\text{O}_7$ (potassium dichromate) ke acidic medium mein oxidize hokar Ethanoic acid deta hai. [Q572, Q574, Q559]
- Secondary alcohol $\rightarrow$ Ketone; Tertiary alcohol $\rightarrow$ resistant to oxidation (strong oxidizing agents se cleavage). [Q572, Q574]

9. Aldehydes & Ketones: Properties, Tests & Differences

- Aldehyde ($-\text{CHO}$) = carbonyl group terminal carbon par; general formula $\text{R}-\text{CHO}$; examples: Methanal HCHO , Ethanal CH_3CHO . [Q576, Q578, Q580]
- Ketone ($>\text{C}=\text{O}$) = carbonyl group middle carbon par; general formula $\text{R}-\text{CO}-\text{R}'$; examples: Propanone CH_3COCH_3 , Butanone $\text{CH}_3\text{COC}_2\text{H}_5$. [Q578, Q580]
- Oxidation: Aldehyde $\rightarrow$ Carboxylic acid (easily oxidized; reducing agent); Ketone $\rightarrow$ resistant to oxidation. [Q580, Q582]

- Reduction (H_2/Ni): Aldehyde $\rightarrow$ Primary alcohol; Ketone $\rightarrow$ Secondary alcohol. [Q582]
- Tollen's test: Aldehyde silver mirror deta hai; ketone nahi. $RCHO + 2[Ag(NH_3)_2]^+ + 3OH^- \rightarrow RCOO^- + 2Ag\downarrow + 4NH_3 + 2H_2O$. [Q584]
- Fehling's test: Aldehyde red Cu_2O precipitate deta hai; ketone nahi. $RCHO + 2Cu^{2+} + 5OH^- \rightarrow RCOO^- + Cu_2O\downarrow + 3H_2O$. [Q584]

10. Carboxylic Acids: Properties, Reactions & Uses

- Carboxylic acid ($-COOH$) = weak acid; examples: Methanoic acid $HCOOH$, Ethanoic acid CH_3COOH . [Q586, Q588, Q590]
- Ethanoic acid = acetic acid; vinegar mein 5–8% solution; carboxylic acids mein sabse high boiling point acetic acid ka hota hai (given options mein). [Q588, Q539]
- Reactions: With metals — $2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2\uparrow$. With bases — $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$. With carbonates — $2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + H_2O + CO_2\uparrow$. [Q590, Q592]
- Esterification: Carboxylic acid + Alcohol $\rightarrow$ Ester + Water (conc. H_2SO_4 catalyst, reversible). $CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O$; sweet smell. [Q592, Q594]
- Saponification: Ester + $NaOH/KOH \rightarrow$ Salt of fatty acid (soap) + Alcohol; Soaps sodium/potassium salts of long chain Carboxylic acids hote hain. [Q594, Q596, Q591]
- Carbonic acid soft drinks mein fizz add karne ke liye use hoti hai (CO_2 dissolved in water). [Q581]

11. Esters: Formation, Hydrolysis & Uses

- Ester formula: $R-COO-R'$; derived from carboxylic acid + alcohol; $CH_3COOC_2H_5$ ek ester ka example hai. [Q594, Q596, Q598, Q583]
- Naming: alkyl group from alcohol + acid name with -oate suffix; $CH_3COOC_2H_5$ = Ethyl ethanoate. [Q598]
- Acid hydrolysis: Ester + H_2O (H^+ catalyst) $\rightarrow$ Carboxylic acid + Alcohol; Ester acid ya base ki presence mein reactant karke wapas alcohol aur carboxylic acid deta hai. [Q596, Q626]
- Alkaline hydrolysis (Saponification): Ester + $NaOH \rightarrow$ Sodium salt of acid + Alcohol. [Q596]
- Uses: perfumes, flavors, solvents, plasticizers. [Q598]
- Glycerol (glycerin) ka IUPAC name Propane-1,2,3-triol hai (3 $-OH$ groups). [Q585]

12. Organic Compounds: Sources, Preparation & Daily Life

- Ethanol: fermentation (yeast + glucose, anaerobic: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$ — yeh anaerobic respiration in yeasts se hota hai); synthesis (ethene hydration). [Q600, Q602, Q587]
- Ethanol ka industrial-use version drinking ke liye unfit banane ke liye Methanol (poisonous) se adulterate/denature kiya jaata hai. [Q630]
- Ethanoic acid: ethanol oxidation; oxidation of acetaldehyde. [Q602, Q604]
- Ethanol se ethene banane ke liye conc. H_2SO_4 ek dehydrating agent ke roop mein use hota hai — 443 K par excess conc. H_2SO_4 ke saath heat karne par $CH_2=CH_2$ (ethene) product milta hai. [Q606, Q607, Q553]
- Methane: natural gas, marsh gas, biogas; anaerobic decomposition. [Q604, Q606]
- Ethene: ethanol dehydration (conc. H_2SO_4 , $170^\circ C$); cracking of petroleum; plastics banane mein use hoti hai (jaise Polythene ke liye). [Q606, Q579]
- Acetylene (Ethyne): calcium carbide + water $\rightarrow CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$. [Q608]
- Important daily life compounds: Methanol (CH_3OH) = wood spirit, poisonous, solvent, denaturant. Formaldehyde ($HCHO$) = preservative (biological specimens), disinfectant, plastics (Bakelite). Acetone (CH_3COCH_3) = nail polish

remover, solvent. Chloroform (CHCl_3) = solvent; historically anaesthetic; air mein phosgene (COCl_2) toxic gas banata hai. Carbon tetrachloride (CCl_4) = solvent; fire extinguisher; ozone depleting. Urea (NH_2CONH_2) = fertilizer; first organic compound synthesized from inorganic (Wöhler, 1828). [Q610]

- Ethyl alcohol (ethanol) ka use fuel (spirit lamps), alcoholic beverage, aur solvent — teenon roop mein hota hai. [Q599]
- Ethyl alcohol hospitals mein wounds aur syringes sterilize karne ke liye antiseptic ke roop mein use hoti hai. [Q595]
- Bahut zyada ethanol consume karne se metabolic processes slow ho jaate hain aur central nervous system depress hota hai. [Q642]
- Ethanol ($\text{C}_2\text{H}_5\text{OH}$ / $\text{C}_2\text{H}_6\text{O}$) ke ek molecule mein 6 hydrogen atoms hote hain. [Q621]
- Propane gas ka use illuminating gas (roshni ke liye) ke roop mein hota hai, cooking aur heating mein bhi. [Q616]
- Ethylene fruit ripening mein use hota hai; ethyne welding mein. [Q606]
- Branched chain ka boiling point lower hota hai straight chain se (surface area kam). [Q604]
- Sugarcane mein Glucose aur Fructose (sucrose banane wale monosaccharides) hote hain. [Q593]
- Glucose ek 6-carbon molecule hota hai ($\text{C}_6\text{H}_{12}\text{O}_6$). [Q637]
- Covalent compounds (jaise organic compounds) poor electricity conductors hote hain, electron sharing se bante hain, aur inka melting/boiling point low hota hai — ye sab properties covalent compounds ki hain. [Q597]

13. Petroleum Refining & Fractional Distillation

- Petroleum = crude oil; mixture of hydrocarbons; fossil fuel. [Q611, Q613, Q615]
- Fractional distillation = different boiling points par fractions alag karte hain: Petroleum gas (LPG) — fuel, cooking. Petrol/Gasoline — motor fuel, dry cleaning. Kerosene — jet fuel, stoves, lamps. Diesel — buses, trucks, trains. Lubricating oil — machinery, grease. Paraffin wax — candles, ointments. Bitumen/Asphalt — roads, roofing. [Q613, Q615, Q617]
- Higher fractions ka molecular weight aur viscosity zyada hoti hai. [Q615]
- Cracking = heavy fractions ko lighter fractions (petrol) mein convert karna; heat + catalyst. [Q617]

14. Coal & Its Products

- Coal = fossil fuel; carbonized plant matter; types: peat, lignite, bituminous, anthracite. [Q618, Q620, Q622]
- Destructive distillation of coal = coal ko absence of air mein heat karna; products: Coke, Coal tar, Coal gas, Ammoniacal liquor. [Q620, Q622]
- Coke = porous, hard; blast furnace mein reducing agent; domestic fuel. [Q622]
- Coal tar = black thick liquid; fractional distillation se benzene, toluene, phenol, naphthalene, anthracene milta hain. [Q622, Q624]
- Coal gas = fuel (H_2 , CH_4 , CO mixture); historically street lighting. [Q624]
- Ammoniacal liquor = fertilizers (ammonium salts). [Q624]

15. Natural Gas, LPG & CNG

- Natural gas mainly methane (CH_4) hota hai; CNG = Compressed Natural Gas. [Q625, Q627, Q629]
- LPG (Liquefied Petroleum Gas) = mixture of butane + propane; liquid form mein store hota hai cylinders mein; domestic fuel. [Q629, Q631]
- CNG vehicles mein use hota hai; pollution petrol/diesel se kam. [Q629]

- Natural gas ka main advantage = clean burning; less pollution. [Q631]

16. Polymers: Natural vs Synthetic, Thermoplastic vs Thermosetting

- Polymer = large molecule; repeating monomer units. [Q632, Q634, Q636]
- Natural polymers: Cotton (cellulose), Silk (protein), Wool (protein), Natural rubber (isoprene), Starch. [Q634, Q636]
- Synthetic polymers: Polythene, PVC, Nylon, Polyester, Teflon, Bakelite. [Q636, Q638]
- Thermoplastics = heat par soft, reshape ho sakte hain, recycle ho sakte hain; examples: Polythene, PVC, Polystyrene, Teflon. [Q636]
- Thermosetting plastics = ek baar set hone ke baad permanently hard; examples: Bakelite, Melamine, Polyester resin. [Q638]
- Addition polymers: Monomer directly add hote hain bina by-product ke; Polythene (ethene), PVC (vinyl chloride), Teflon (tetrafluoroethene). [Q638]
- Condensation polymers: Monomer ke saath by-product (water, HCl) nikalta hai; Nylon, Polyester, Bakelite. [Q638]
- Nylon = polyamide; socks, ropes, parachutes; Terylene/Dacron = polyester; fabrics, bottles; Bakelite = phenol-formaldehyde resin; electrical switches, handles. [Q638, Q640]

17. Rubber: Natural & Synthetic

- Natural rubber = rubber tree ka latex; polymer of isoprene (2-methyl-1,3-butadiene); soft aur sticky. [Q641, Q643]
- Vulcanization = natural rubber ke saath 3–5% sulphur heat karna (373–415 K); cross-linking se rubber hard, elastic, durable hota hai; Charles Goodyear ne discover kiya. [Q641, Q643]
- Synthetic rubber: Neoprene (chloroprene polymer); oil-resistant; hoses, gaskets. Buna-S (Styrene-butadiene rubber); tyres. Buna-N (Nitrile rubber); oil/fuel resistant. [Q643]
- Vulcanized rubber ka use: tyres, shoes, belts, hoses. [Q643]
- Carbon ka Fullerene allotrope geodesic globe (soccer-ball jaisi) shape mein hota hai. [Q635]

18. Environmental Impact & Non-biodegradable Substances

- Plastics non-biodegradable hain; soil aur water pollution karte hain. [Q643]
- Plastic burning se toxic gases (dioxins, furans) release hote hain. [Q643]
- 4R principle: Reduce, Reuse, Recycle, Recover. [Q643]
- CFCs (Chlorofluorocarbons) ozone layer deplete karte hain; replaced by HFCs. [Q643]
- Biodegradable plastics research chal rahi hai (starch-based, PLA). [Q643]

19. ⚠ Traps / Key Differences Summary (Q532–Q643)

- ⚠ Trap: Homologous series mein successive members ka molecular mass difference 14 u hota hai ($\text{CH}_2 = 12+2$), 12 ya 16 nahi. [Q532]
- ⚠ Trap: Same molecular formula wale alag functional groups same homologous series mein nahi hain — C_2H_6 (alkane), C_2H_4 (alkene), C_2H_2 (alkyne) alag hain. [Q533]
- ⚠ Trap: Alkane = single bond $\text{C}_n\text{H}_{2n+2}$; Alkene = double bond C_nH_{2n} ; Alkyne = triple bond $\text{C}_n\text{H}_{2n-2}$; Cycloalkane = ring C_nH_{2n} — sab alag formulas hain. [Q534, Q536, Q538, Q540]

- ⚠ Trap: Chain isomerism mein skeleton alag; Position isomerism mein functional group ki position alag; Functional group isomerism mein functional group hi alag — teenon alag hain. [Q552, Q554]
- ⚠ Trap: Saturated = clean flame; Unsaturated = sooty flame — carbon:hydrogen ratio se decide hota hai. [Q556, Q558, Q560]
- ⚠ Trap: Addition reaction sirf unsaturated compounds mein; Substitution sirf saturated compounds mein — confuse mat hona. [Q561, Q566]
- ⚠ Trap: Primary alcohol oxidize hokar aldehyde phir acid banata hai; Secondary ketone banata hai; Tertiary resistant hota hai. [Q572, Q574]
- ⚠ Trap: Aldehyde terminal carbonyl ($-\text{CHO}$); Ketone middle carbonyl ($>\text{C}=\text{O}$) — Tollen's aur Fehling's test aldehyde ke liye positive, ketone ke liye negative. [Q576, Q578, Q580, Q584]
- ⚠ Trap: Esterification reversible hai; conc. H_2SO_4 catalyst + dehydrating agent hai jo water absorb karke equilibrium ester ki taraf shift karta hai. [Q592, Q594]
- ⚠ Trap: Methanol (CH_3OH) poisonous hai; Ethanol ($\text{C}_2\text{H}_5\text{OH}$) drinkable (controlled amount) — confuse mat hona; isiliye industrial ethanol ko methanol se adulterate karke unfit banaya jaata hai. [Q610, Q630]
- ⚠ Trap: Chloroform air mein phosgene (COCl_2) toxic gas banata hai; isiliye modern anaesthesia mein use nahi hota. [Q610]
- ⚠ Trap: Urea first organic compound tha jo laboratory mein inorganic se synthesize kiya gaya (Wöhler, 1828) — vital force theory ko disprove kiya. [Q610]
- ⚠ Trap: Petroleum fractional distillation mein petrol pehle nikalta hai (low BP); asphalt/bitumen sabse last mein (high BP). [Q613, Q615, Q617]
- ⚠ Trap: LPG = butane + propane; CNG = mainly methane — donon alag hain. [Q629, Q631]
- ⚠ Trap: Thermoplastic heat par melt ho sakta hai, reshape aur recycle ho sakta hai; Thermosetting ek baar set hua toh permanently hard — confuse mat hona. [Q636, Q638]
- ⚠ Trap: Addition polymerization mein koi by-product nahi; Condensation polymerization mein by-product (water/HCl) nikalta hai. [Q638]
- ⚠ Trap: Vulcanization mein sulphur add karte hain cross-linking ke liye; natural rubber soft/sticky hota hai, vulcanized rubber hard/elastic hota hai. [Q641, Q643]
- ⚠ Trap: Neoprene oil-resistant hai; Buna-S tyres mein; Buna-N fuel-resistant — alag-alag uses hain. [Q643]
- ⚠ Trap: Branched chain ka boiling point lower hota hai straight chain se; exam mein order pucha jaata hai. [Q604]
- ⚠ Trap: Bond length order — triple bond (Ethyne) sabse chhoti, phir double bond (Ethene), phir single bond (Ethane) sabse badi. [Q609]
- ⚠ Trap: Hydrogenation ka catalyst hamesha Nickel (Ni) hota hai, Platinum nahi (is context mein). [Q612, Q639]

20. Numerical: Molecular Mass Difference in Homologous Series

- Homologous series mein har successive member ka difference CH_2 group hota hai. [Q644, Q646]
- CH_2 group ka molecular mass = $\text{C} (12) + \text{H}_2 (2) = 14 \text{ u}$. [Q646]
- Isiliye do kramagat members ka molecular mass hamesha 14 u se differ karta hai. [Q646]
- Example: CH_4 (16 u) aur C_2H_6 (30 u) ka difference = $30 - 16 = 14 \text{ u}$. [Q646]
- Example: C_2H_6 (30 u) aur C_3H_8 (44 u) ka difference = $44 - 30 = 14 \text{ u}$. [Q646]
- Ye rule sabhi homologous series ke liye valid hai — chahe alkanes ho, alkenes, alcohols, ya carboxylic acids. [Q646]

21. Numerical: Calculations — Alcohols & Hydrocarbons

- Methanol (CH_3OH) ka molecular mass = $12 + 3(1) + 16 + 1 = 32 \text{ u}$. [Q645]

- Ethanol ($\text{C}_2\text{H}_5\text{OH}$) ka molecular mass = $2(12) + 5(1) + 16 + 1 = 46$ u. [Q645]
- Methanol aur ethanol ka difference = $46 - 32 = 14$ u (CH_2 group). [Q645]
- Propane (C_3H_8) ka molecular mass = $3(12) + 8(1) = 44$ u. [Q644]
- Pentane (C_5H_{12}) ka molecular mass = $5(12) + 12(1) = 72$ u. [Q644]
- C_3H_8 aur C_5H_{12} ka difference = $72 - 44 = 28$ u (do CH_2 groups ka difference). [Q644]

22. Numerical: Formula Cheat Sheet

- CH_2 group mass: $12 + 2 = 14$ u.
- Molecular mass of alkane: $12n + (2n+2) = 14n + 2$.
- Molecular mass of alkene: $12n + 2n = 14n$.
- Molecular mass of alcohol: $12n + (2n+1) + 16 + 1 = 14n + 18$.
- Difference between successive members: 14 u.

23. ⚠ Traps / Key Differences Summary (Numerical Q644–Q646)

- ⚠ Trap: C_3H_6 alkene (propene) hai jiska mass 42 u hai; C_3H_8 alkane (propane) hai jiska mass 44 u hai — 2 u ka difference hai, confuse mat hona. [Q644]
- ⚠ Trap: Do members ke beech agar ek member gap ho toh difference 28 u (2×14) hoga, 14 nahi. [Q644]
- ⚠ Trap: Alcohols mein bhi same 14 u ka difference hota hai kyunki OH group common rehta hai aur carbon chain CH_2 se badhti hai. [Q645]
- ⚠ Trap: Atomic masses yaad rakhna: C = 12, H = 1, O = 16 — calculation fast karni padti hai exam mein. [Q644, Q645]